MLA02005 – Fundamentals of Machine Learning.

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BTech. CSE. CS
Slot - A.

ASSIGNMENT – II

SUPERVISED LEARNING

Task 1. Linear Regression

Given, $x = 1, 2, 3, 4, 5$

$y = 2.1, 3.9, 6.2, 7.8, 10.1$

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(a) Regression equation,

Using the normal equations :

$$w_1 = \frac{5(110.2) - 15(30.1)}{5(55) - 15^2} = 1.99$$

$$w_0 = \bar{y} - w_1\bar{x}$$

$$= 6.02 - 1.99(3) = 0.05$$

$$\therefore \quad y = 0.05 + 1.99x.$$

(b) Prediction at $x = 6$,

$$y = 0.05 + 1.99(6) = 11.99$$

(c) Evaluation,

$RSS = 0.107$, $MSE = \frac{0.107}{5} = 0.0214$

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$$R^2 = 1 - \frac{0.107}{39.708} = 0.9973$$

(d) Reliability : The fit is excellent, but predicting at $x = 6$ is extrapolation beyond the given data. Hence, the prediction may be less reliable.

Task 2. Ridge Regression.

Given, $S_{xx} = 10$

$S_{xy} = 19.9$

$\lambda = 5$

Ordinary least-squares slopes:

$$w_{LS} = \frac{19.9}{10} = 1.99$$

Ridge slope: $w_R = \frac{19.9}{10+5} = 1.3267$

2 percentage shrinkage: $\frac{1.99 - 1.3267}{1.99} \times 100 = 33.33\%$.

Effect: Increasing λ reduces variance and overfitting but increases the bias.

Task 3. Naive Bayes.

(a) Priors,

$$P(Spam) = P(Ham) = \frac{5}{10} = 0.5$$

Conditional probability	Spam	Ham
P(Offer = Y)	0.8	0.2
P(Offer = N)	0.2	0.8
P(Link = Y)	0.8	0.2
P(Link = N)	0.2	0.8

(b) Offer = Y, Link = Y (without smoothing)

$$P(spam \mid YY) = \frac{0.5(0.8)(0.8)}{0.5(0.8)(0.8) + 0.5(0.2)(0.2)}$$

$$P(\text{Spam} \mid YY) = 94.12\%.$$

Classification : Spam.

(c) With Laplace smoothing, $\alpha = 1$.

$$P(\text{Spam} \mid YY) = \frac{25}{25+4} = 86.21\%.$$

Classification : Spam.

(d) Offer = N, Link = N (with smoothing)

$$P(\text{Spam} \mid NN) = \frac{4}{29} = 13.79\%.$$

Classification : Ham.

5 The independence assumption simplifies calculations but may not always hold. Smoothing prevents zero probabilities and reduces extreme predictions.

Task4. Logistic Regression.

Initially, $w_1 = w_2 = b = 0$, learning rate $\eta = 0.5$

Both initial predicted probabilities are 0.5

Gradient of mean cross-entropy loss :

$$\frac{\partial L}{\partial w_1} = 0.25, \quad \frac{\partial L}{\partial w_2} = -0.25, \quad \frac{\partial L}{\partial b} = 0$$

Updated parameters,

$$w_1 = 0 - 0.5(0.25) = -0.125$$

$$w_2 = 0 - 0.5(-0.25) = 0.125, \quad b = 0$$

Updated probabilities :

Sample	Initial	Updated.
(1,2)	0.5000	0.5312
(2,1)	0.5000	0.4688

$L_{before} = 0.6931$, $L_{after} \approx 0.6320$

Result : The loss decreases after the update, improving predictions.

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Task5. Decision Tree - Entropy and Gini.

Given, 6 Yes , 4 No ; Split into $a_1 = (4Y, 0N)$

$a_2 = (2Y, 4N)$

Parent entropy , $H(S) = -0.6\log_2 0.6 - 0.4\log_2 0.4 = 0.971$

Weighted child entropy,

$$H_{split} = \frac{4}{10}(0) + \frac{6}{10}(0.9183) = 0.551$$

Information gain,

$$IG = 0.971 - 0.551 = 0.420$$

Gini decrease ,

$$G(S) = 1 - (0.6)^2 + (0.4)^2 = 0.48$$

$$G_{split} = 0.6(0.4444) = 0.2667$$

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Gini decrease $= 0.2133$

The split is worthwhile because both entropy and Gini impurity decreases.

Task6. Cost - Complexity Pruning.

Subtree : 3 errors, 4 leaves. Single leaf : 8 errors, 1 leaf.

$N = 100$

$$R_\alpha(T_s) = 0.03 + 4\alpha$$

$$R_\alpha(T_l) = 0.08 + \alpha$$

Equating the costs,

$$0.08 + 4\alpha = 0.08 + \alpha$$

$$\alpha = 0.01667$$

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→ For $\alpha = 0.01$ · Substree cost $= 0.07$, leaf cost $= 0.09$. Keep subtree

→ For $\alpha = 0.02$: subtree cost $= 0.11$, leaf cost $= 0.10$. Prune subtree

Task7. Neural Network and Backpropagation.

Given,

$x = (1, 2)$, $t = 1$, $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$, $\eta = 0.5$

Forward pass :

$$h = \sigma(0.4 + 0.6) = 0.7311$$

$$\hat{y} = \sigma(0.5h) = 0.5904$$

$$E = \frac{1}{2}(1 - 0.5904)^2 = 0.0839.$$

Backpropagation :

$$\delta_o = (\hat{y} - t)\hat{y}(1 - \hat{y}) = -0.09906$$

$$\delta_h = \delta_o v h(1 - h) = -0.00974$$

Gradient,

$$\frac{\partial E}{\partial v} = -0.07242$$

$$\frac{\partial E}{\partial w_1} = -0.00974, \quad \frac{\partial E}{\partial w_2} = -0.01948.$$

updated weights,

$$v = 0.58621, \quad w_1 = 0.40487, \quad w_2 = 0.30974$$

Now loss:

$$\text{Error} \approx 0.0811.$$

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Result: The loss decreases after updating the weights.

Task8. Support Vector Machine:

Given,

$$w = (0.5, 0.5), \quad b = -2$$

$$f(x) = 0.5x_1 + 0.5x_2 - 2$$

(a) Support vectors,

$$f(3,3) = 1, \quad f(1,1) = -1.$$

Both lie on the margin hyperplanes.

$$\text{Margin width} = \frac{2}{\|w\|} = \frac{2}{\sqrt{0.5}} \approx 2.8284.$$

(b) Classification and distance,

$$d = \frac{|f(x)|}{\|w\|}$$

point	$f(x)$	Class	Distance.
(4,2)	1	+1	1.4142
(1,2)	-0.5	-1	0.7071

(c) Hinge loss,

$$L = \max(0, 1 - y f(x))$$

For (1,2), $y=-1$: $L=0.5$

For (2,2), ~~$y=+1$~~ : $L=1$

4 Result : Both points violate the margin condition.

Task9. Ensemble Learning.

Each independent classifier has accuracy $p=0.7$,

For 3 classifiers,

$$P_3 = 3(0.7)^2(0.3)+(0.7)^3$$

$$P_3 = 78.4\%.$$

For 5 classifiers,

$$P_5 = \left(\frac{5}{3}\right)(0.7)^3(0.3)^2 + \left(\frac{5}{4}\right)(0.7)^4(0.3)+(0.7)^5$$

$$P_5 = 83.692\%.$$

3 Adding classifiers improves accuracy when their errors are independent. Strongly correleated errors reduce the benefit of the majority voting.

Task10. AdaBoost

Five samples have equal initial weights 0.2, samples 2 and 5 are misclassified.

Weighted error :

$$e = \frac{2}{5} = 0.4$$

Learner weight :

$$\alpha = \frac{1}{2}\ln\frac{1-e}{e} = \frac{1}{2}\ln\frac{0.6}{0.4} = 0.2027.$$

Updated weights :

Correctly classified samples,

$$0.2e^{-0.2027} \approx 0.1633$$

Misclassified samples,

$$0.2e^{0.2027} \approx 0.2449.$$

After normalisation,

Samples	Normalised weight.
1, 3, 4	0.1667 each
2, 5	0.2500 each

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AdaBoost increase the weights of misclassified samples. so that next learner focuses on difficult examples.

Task 11. Linear Discriminant Analysis.

Given,

$$\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}, \quad \mu_1 = (1,2)^T, \quad \mu_2 = (3,1)^T$$

(a) Discriminant direction,

$$\Sigma^{-1} = \begin{bmatrix} 4/7 & -2/7 \\ -2/7 & 8/7 \end{bmatrix}$$

$$w = \Sigma^{-1}(\mu_2 - \mu_1)$$

$$w = (10/7, -12/7)^T$$

For equal priors :

$$c = \frac{1}{2}(\mu_1 + \mu_2)^T w = 0.2857$$

(b) Classify $x = (2.5, 1)$

$$w^T x = \frac{13}{7} = 1.8571.$$

Since $1.8571 > 0.2857$

Classify as w_2.

(c) Unequal priors,

For $P(w_1) = 0.8$, $P(w_2) = 0.2$

$$C' = 0.2857 + \ln \frac{0.8}{0.2} = 1.6710.$$

Since, $1.8571 > 1.6710$ the class remains w_2

Task12 - Bank Loan Classification.

Given, 1000 loans and 100 actual defaults.

(a) Evaluation metrics,

$$\text{Accuracy} = \frac{TP + TN}{1000}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2TP}{2TP + FP + FN}$$

Metric	Model A	Model B	Model C
Accuracy	88%	80.5%	93%
Precision	43.75%	32.08%	66.67%
Recall	70%	85%	60%
F1-score	53.85%	46.58%	63.16%

(b) Total expected cost,

$$Cost = FN(₹50,000) + FP(₹5,000)$$

Model	Calculation	Total cost
A	30(50000)+90(5000)	1950000
B	15(50000)+180(5000)	1650000
C	40(50000)+30(5000)	2150000

(c) Best model,

Model B, is preferred based on cost because its total cost is lowest at ₹16,50,000. It also has the highest recall, so it misses the fewer defaulters.

Model C has the highest accuracy and F1-score but the highest score cost because it misses 40 defaulters.

(d) Improvements,

1. Tune the classification threshold to reduce costly missed defaulters.
2. Use cost-sensitive learning to give greater importance to missed defaulters.
3. Use cross-validation and a separate test set to prevent overfitting.

Conclusion: Model B is the best choice under the given cost assumptions.